

A Novel Method for Detection and Tracking of Low-Slow-Small Drones Based on Computer Vision

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Abstract—To enhance the monitoring capability of low-slow-small (LSS) drones, a novel method with higher precision and stability for the real-time detection and tracking of LSS drones based on computer vision (CV) is proposed. By integrating the feature pyramid structure, deformable convolution, attention mechanism, and online hard example mining into the faster R-CNN framework, an advanced CV detection model named Drone-Net is developed. This model better balances speed and accuracy in drone tracking application scenarios and enhances performance for small targets, multi-scale, and similar interference situations. To address the limitations of traditional angle-based PID control methods, such as hysteresis and error accumulation, a velocity adaptive tracking control algorithm based on multi-level judgment is designed. Experimental results show that the strategies introduced into Drone-Net effectively improve its general performance, with the mean Average Precision (mAP) at 69.1. The two-axis gimbal system built according to the proposed method performs stably and efficiently in drone tracking.

Keywords—computer vision; drone monitoring; velocity adaptive tracking; two-axis gimbal control

I. Introduction

Low-Slow-Small (LSS) drones are characterized by their low-altitude flight, slow speed, and small in size and weight. As technology continues to advance and prices steadily decrease, LSS drones have found extensive applications in both military and civilian sectors. While LSS drones bring significant societal benefits, various low-altitude security risks are introduced, such as uncontrolled or collisions in mid-air, unauthorized intrusions, and information leakage. In the military sector, Integrated Air Defense and Missile Defense (IADM) systems struggle to effectively detect LSS drones [1]. In the civilian sector, there is also a lack of effective detection methods. Developing an efficient real-time monitoring and tracking integrated system for LSS drones is now an urgent necessity.

At present, based on detection methods and positioning techniques, the LSS drone monitoring technologies can be categorized into acoustic monitoring technology, radar

monitoring technology, radio-passive monitoring technology, and optical-electronic monitoring technology [2]. Among these, acoustic monitoring technology relies on the time and intensity of received sound signals to determine the drone's position. However, this method is highly susceptible to environmental factors such as noise and temperature. Radar monitoring technology calculates the target's distance, speed, and azimuth by analyzing the emitted electromagnetic waves and received echoes. This method is suitable for detecting high-altitude, fast-moving targets with a relatively large radar-cross-sectional area (RCS), but not effective for LSS drones with low metal content and small RCS. Radio-passive monitoring technology detects drones by their working frequency band of remote-control signals. This method will be ineffective when facing drones with silent radios. Optical-electronic monitoring technology mostly employs computer vision to detect drones by analyzing the real-time images in low-altitude airspace. This method has the advantages of strong anti-interference, intuitive results, and flexible deployment stations.

The drone detection and tracking method that combines Computer Vision (CV) with a two-axis gimbal is widely adopted [3]. The hardware platform for this approach generally consists of a high-performance embedded processor and a two-axis gimbal equipped with a camera. Firstly, the collected low-altitude images are processed using a CV detection model to extract the drone's coordinate information. Subsequently, this coordinate data is utilized to compute the control signals needed for operating the gimbal. Lastly, the two-axis gimbal rotates to track the target drone. In recent years, there have been many studies based on this method. For instance, Xiufang Shi et al. [4] proposed an anti-drone system with multiple surveillance technologies. The system employs feature extraction techniques and a Support Vector Machines algorithm for image detection. Subsequently, it uses frame differencing methods and PID algorithms to control the gimbal rotation for real-time tracking of drones. Bor-Horng Sheu et al. [5] designed a two-axis gimbal with an image recognition function to detect and track drones. The system can track intruding drones within detection range through a nine-axis sensor and laser rangefinder. Many current methods for drone detection and tracking rely on CV technology. While these

methods have addressed the issue of detecting small targets, they often neglect problems like interference from similar targets, target occlusion, and handling multi-scale targets [6]. Moreover, most of these approaches employ traditional PID control for two-axis gimbals, which is less effective for tracking drones with unpredictable and complex flight paths [7].

In this study, we integrate the feature pyramid structure, deformable convolution, attention mechanism, and online hard example mining into the faster R-CNN [8] framework to develop an advanced CV detection model optimized for LSS drones. This model better balances speed and accuracy in drone tracking application scenarios, enhancing performance for small targets, multi-scale, and similar interference situations. To address the limitations of the traditional angle-based PID control methods, such as hysteresis and error accumulation, we introduce a velocity adaptive tracking control algorithm based on multi-level judgment. On this basis, a method with higher precision and stability for the real-time detection and tracking of LSS drones based on CV is proposed.

II. Drone-Net model

In the system for detecting and tracking LSS drones, the performance is heavily reliant on the CV model. The system's reliability hinges on detection accuracy, while its real-time performance is dictated by detection speed. However, the trade-off between detection accuracy and detection speed poses a common challenge in object detection algorithms. The object detection algorithms with anchor boxes exhibit higher accuracy performance compared to those without anchor boxes. It can be divided into two categories: one-stage detection algorithms, such as SSD [9] and YOLO [10], and two-stage detection algorithms, like Faster R-CNN [8] and Mask R-CNN [11]. Two-stage detection algorithms, after the feature extraction, require a region proposal network to generate suggested areas for secondary regression prediction. One-stage detection algorithms are based on pre-generated random prior boxes and directly perform classification and regression. These algorithms are simpler and require less computational power. While they are faster, they generally yield lower detection accuracy compared to two-stage algorithms.

Nonetheless, given the advancements in computing power and the specific application context of this paper, the slower detection speed of two-stage algorithms becomes less of a concern. Furthermore, in scenarios featuring low-altitude drones with small, multi-scale targets and similar interference, two-stage detection models can deliver superior performance. In this section, we will develop the Drone-Net drone detection model based on the two-stage target detection algorithm: Faster R-CNN.

A. Model design

To enhance the detection accuracy and stability in scenarios with small, multi-scale targets and similar interference, while also considering real-time performance, we have developed the Drone-Net drone detection model. The core technology of this model is integrating the feature pyramid structure [12], deformable convolution [13], attention mechanism [14], and

online hard example mining [15] into the Faster R-CNN framework ("Resnet-101" as the feature extraction network).

1) *Feature pyramid structure*: Incorporating the feature pyramid structure into the Drone Net model effectively addresses the issues of detail loss or redundancy during the downsampling of small target information. The specific architecture is illustrated in Fig. 1 and comprises three components: feature downsampling, feature upsampling, and cross-addition. The feature downsampling process employs convolution and downsampling procedures to derive multi-level feature maps from a low-altitude image, capturing both intricate details in the shallow layers and high-level semantic information in the deeper layers. The feature upsampling process builds upon these downsampled feature maps. By continuously expanding the dimensions, the expressiveness of each feature map is enhanced. The cross-addition process involves stacking feature maps produced from both upsampling and downsampling on a layer-by-layer basis. This not only recovers information lost during the upsampling phase but also minimizes the dispersion of image features.

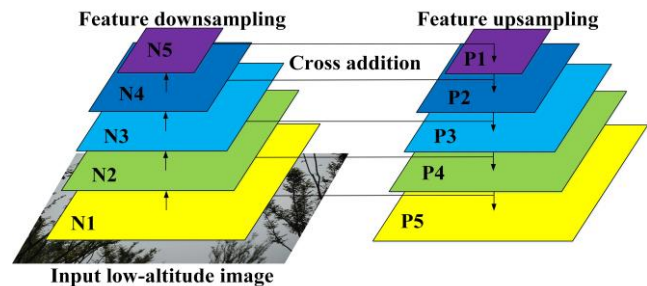


Figure 1. The feature pyramid structure in Drone-Net.

2) *Deformable convolution*: Compared to standard convolution, deformable convolution introduces offsets to break the sampling limitations of regular regions in feature maps. This effectively addresses the difficulty of standard convolution in extracting useful features from targets with deformations. As shown in Fig. 2, the convolution kernel of deformable convolution contains a series of two-dimensional weights (or offsets), which can be auto-adjusted during the backpropagation process. In the Drone-Net model featuring a ResNet101-based feature extraction network, standard convolution layers from the third to the fifth layer are replaced with deformable convolutions. This modification is particularly beneficial when the feature map dimensions are smaller, as the computational overhead of calculating offsets has a relatively low impact on detection speed, while still maintaining adaptability to variations in target scale.

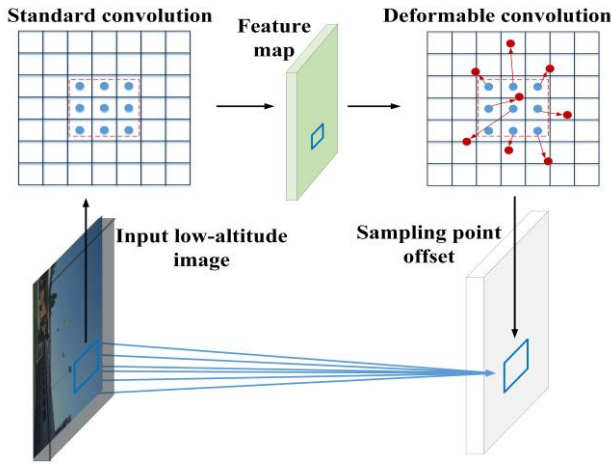


Figure 2. The deformable convolution in Drone-Net.

3) *Attention mechanism*: Incorporating the attention mechanism into Drone-Net enhances feature extraction methods and enriches semantic information, thereby improving the model's capability to handle low-altitude images with complex backgrounds. Additionally, the integration of self-attention can also reduce model complexity and increase detection speed. As depicted in Fig. 3, the input feature map first undergoes a 1×1 convolution to reduce the channel count to one. This is followed by a SoftMax function to fuse it with the original feature map. The fused feature map then passes through another 1×1 convolution, layer normalization, and ReLU activation mapping in the feature dimension to further reduce the channel count, thereby decreasing computational complexity while enhancing feature extraction. Finally, a 1×1 convolution is applied to restore the original channel count. The resulting global context features are added to the original features to yield an attention-focused feature map.

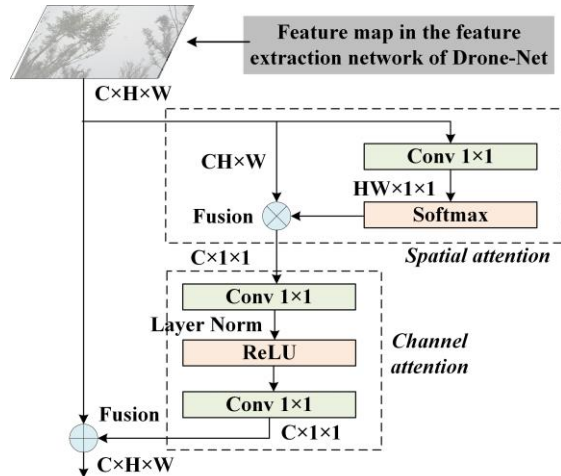


Figure 3. The attention mechanism process in Drone-Net.

4) *Online hard example mining*: The introduction of Online Hard Example Mining (OHEM) enables the Drone-Net model to more effectively learn from difficult samples that are

prone to misclassification. As illustrated in Fig. 4, OHEM is implemented by utilizing the feature map output from the final layer of the convolutional feature extractor on low-altitude images. The Region of Interest (RoI) read-only layer performs forward propagation based on proposed regions to calculate the loss value for each area. Training samples are selected according to the magnitude of their loss values and are assembled into training batches at a positive-to-negative sample ratio of 1:3. These hard examples are then fed into the RoI hard layer for further forward and backward propagation, with parameter learning, carried out via a binary classification regression network.

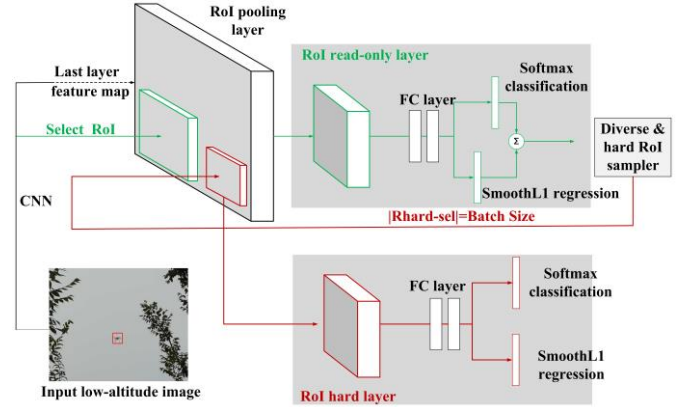


Figure 4. The online hard example mining structure in Drone-Net.

B. Dataset organization

We organize the "Drone-Kite-Bird" (DKB) dataset in accordance with the PASCAL VOC [16] format. Drone raw images collection primarily utilizes USB cameras and smartphones, targeting multi-rotor drones with a wheelbase ranging from 25 cm to 120 cm, and at distances between 10 m and 500 m. To enrich the drone image dataset, data augmentation is performed based on the publicly available USC [17] dataset. To introduce similar distractors, images of kites and flying birds are also collected. The dataset consists of about 16,000 samples, its demonstration samples as shown in Fig. 5. Annotations are completed using Labellmg, and the dataset is split into training and testing sets at a ratio of 8:2.



Figure 5. The demonstration samples from the DKB dataset.

III. Tracking algorithm and system

A. Velocity-adaptive tracking control algorithm

The primary shortcomings of conventional two-axis gimbal control methods employing angle-based PID algorithms include 1) the necessity for recalibration with each scene change; 2) inherent latencies in command transmission and uncertain actuation times for the gimbal, which constrain the control system to update tracking commands only after receiving feedback from the gimbal; 3) difficulties in determining the relative motion between the gimbal and the drone when represented in a static image, thereby resulting in considerable tracking inaccuracies; 4) protracted intervals between consecutive command issuances, heightening the risk of target loss in instances of sudden drone trajectory modifications. To address the problems aforementioned, our study investigates the dynamic layer driving mechanisms of dual-axis gimbals and designs a velocity-adaptive tracking control algorithm based on multi-level judgments. The algorithm offers advantages such as procedural simplicity, rapid response, and high control stability. The specific workflow is illustrated in Fig. 6.

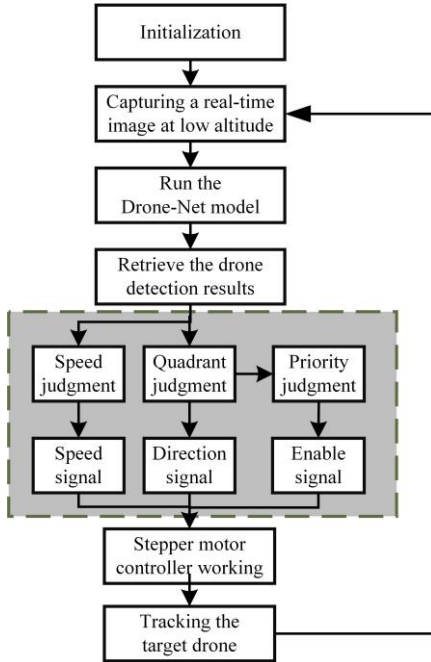


Figure 6. Flow of the tracking algorithm.

Nonetheless, given the advancements in computing power and the specific application context of this paper, the slower detection speed of two-stage algorithms becomes less of a concern. Furthermore, in scenarios featuring low-altitude drones with small, multi-scale targets and similar interference, two-stage detection models can deliver superior performance. In this section, we will develop the Drone-Net drone detection model based on the two-stage target detection algorithm: Faster R-CNN.

After initialization, the Drone-Net model identifies LSS drones in a real-time low-altitude image and produces bounding box data (if exists). A control signal is generated by

calculating the pixel difference between the image center and the predicted drone bounding box center and inputting it into our tracking algorithm. The multi-level decision-making process is shown in Fig. 7. The first level compares projection points on the X-axis and Y-axis of the pixel difference vector with preset velocities V_1 , V_2 , and V_3 to determine the Speed signals. The second level generates the Direction signals based on the quadrant (I~IX) in which the bounding box center resides. The third level gives the Enable signals. If multiple drones are detected, aim to keep as many as possible within the field of view. Finally, these control signals are transmitted to the two-axis gimbal's stepper motor controllers for posture adjustment and drone tracking.

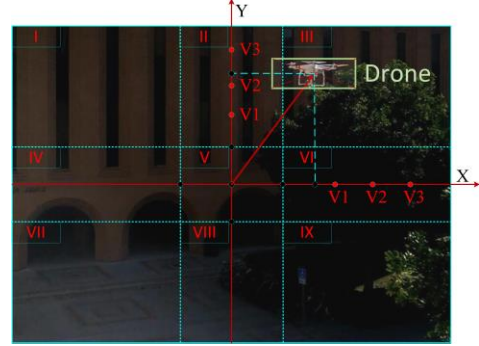


Figure 7. Schematic of the multi-level judgments.

B. Method of achieving the tracking system

We have designed and constructed a detection and tracking system for LSS drones. As shown in Fig. 8, the hardware components of the system primarily consist of a two-axis gimbal, a camera, an embedded processor, and other essential mechanisms and circuitries. The two-axis gimbal features two 57-stepper motors integrated with controllers. In our method, the stepper motors are directly controlled by the Speed, Direction, and Enable signals to drive the gimbal. The camera is the RER VISION 1080P USB-Camera for real-time acquisition of low-altitude images. An NVIDIA Jetson Nano 8GB serves as the embedded processor, running the Drone-Net model under the TensorFlow framework for drone detection. Additionally, it computes the control commands based on the predicted bounding box information to achieve tracking control via the two-axis gimbal.

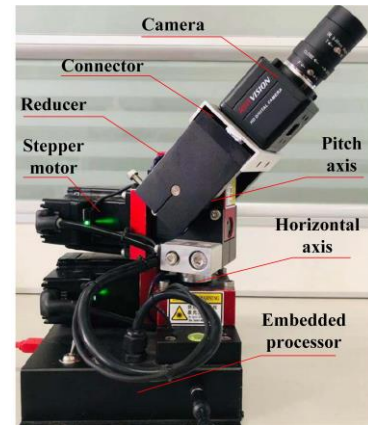


Figure 8. Structure diagram of the tracking system.

IV. Experimental results and discussion

A. Results of the Drone-Net model training

To validate the efficacy of the proposed Drone-Net model, a series of ablation experiments are conducted focusing on the four employed technologies: feature pyramid structure(F), deformable convolutions (D), attention mechanism (A), and online hard example mining (O). The model was trained in an Ubuntu 18.04 environment with TensorFlow framework, using a GeForce RTX 3090 Ti GPU with 48GB of VRAM. During training, we used regularization with a coefficient of 0.0001 and the SGD (Stochastic Gradient Descent) algorithm with a momentum factor of 0.9. A learning rate decay strategy is applied with an initial rate of 0.001, decaying by a factor of 0.1 every five epochs.

The results of model training, listed in Table 1, indicate that the feature pyramid structure effectively increased the detection accuracy for kites and birds, albeit at a 9.7% reduction in detection speed. The deformable convolution slightly decreased the detection accuracy for kites but enhanced the bird detection accuracy by 20%, with a 17.6% decrease in detection speed. The attention mechanism improved the mean Average Precision (mAP) by 1.2 while reducing the detection speed by 3.6%. The OHEM significantly increased the detection accuracy for birds without compromising detection speed. In summary, all the aforementioned techniques contribute to an improvement in the model's general performance.

Table 1. Results of the training experiment.

Model	AP drone	AP kite	AP bird	mAP	FPS
RCNN	91.2	72.2	10.2	58.5	45.2
RCNN +F	91.5	82.1	18.5	64.0	40.8
RCNN +F+D	91.6	81.1	22.2	65.0	33.6
RCNN +F+D+A	91.8	82.9	25.0	66.2	32.4
Drone-Net (↑+O)	91.9	82.5	33.0	69.1	32.4

Some detection results of Drone-Net on difficult samples are given in Fig. 9. In hard scenarios, such as small-target presence, tree occlusions, and kite interferences, Drone-Net consistently exhibits stable accuracy. However, it is prone to missing detections for birds due to their small size.

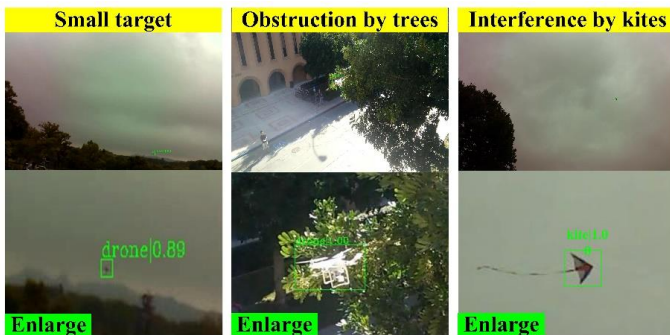


Figure 9. Some drone detection results on difficult samples.

B. Results of LSS drone tracking

The LSS drones had an approximate flight speed of 30 cm/s in this experiment. The V1, V2, and V3 in the velocity-

adaptive tracking control algorithm with settings of 6, 12, and 18 deg/s, respectively. Test scenarios include indoor at 15 m, outdoor at 45 m, and outdoor at 80 m. Conduct drone tracking tests for about one minute in each scenario. Drone tracking videos are captured at a frame rate of 30 FPS, from which a 20-second mid-segment (comprising a total of 1800 frames) is extracted. The probability of the target drone being positioned within the central 5%, 10%, 20%, and 100% areas of the field of view is statistically analyzed across these 1800 frames and introduced as the evaluative criteria for assessing the tracking performance. The statistical results are given in Table 2. The examples of tracking result frames in different scenarios as shown in Fig. 10. The results of the tracking system evaluation substantiate the efficacy of the method proposed in this paper.

Table 2. The statistical results of LSS drone tracking

Scenario	5% area	10% area	20% area	100% area
Indoor 15 m	0.6261	0.8867	0.9628	1.0000
Outdoor 45 m	0.6700	0.8756	0.9561	1.0000
Outdoor 80 m	0.7122	0.9256	0.9922	1.0000

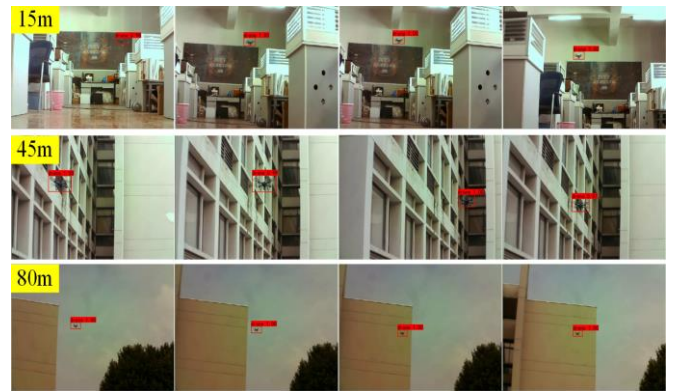


Figure 10. The examples of drone tracking result frames.

V. Conclusions

In conclusion, a method with higher precision and stability for the real-time detection and tracking of LSS drones based on CV is proposed, which could become a new means to improve the monitoring capability of LSS drones. The innovative contributions of this paper can be summarized as the following two points:

1. By integrating the feature pyramid structure, deformable convolution, attention mechanism, and online hard example mining into the faster R-CNN framework, an advanced CV detection model named Drone-Net is developed. Results of the ablation experiments prove these technologies effectively improve its general performance. The accuracy improved by 18% compared to basic faster R-CNN, achieving a mAP of 69.1. This model better balances speed and accuracy in drone tracking application scenarios and enhances performance for small targets, multi-scale, and similar interference situations.

2. To address the limitations of traditional angle-based PID control methods, such as hysteresis and error accumulation, a velocity adaptive tracking control algorithm based on multi-level judgment is designed. Experimental results of the LSS

tracking test show that the two-axis gimbal system built according to the proposed method performs stably and efficiently in drone tracking, with a 100% tracking success rate.

However, it is worth noting that the current Drone-Net model faces challenges in dealing with similar target interference from small flying birds, indicating an avenue for future research.

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References

- [1] Zhou T, Guo J, Wu Q, et al. An unmanned aerial vehicle identification and tracking system based on weakly supervised semantic segmentation technology[J]. *Physical Communication*, 2022, 54: 101758.
- [2] Jiang N, Wang K, Peng X, et al. Anti-UAV: A large multi-modal benchmark for UAV tracking[J]. *arXiv preprint arXiv:2101.08466*, 2021.
- [3] Opromolla R, Fasano G, Accardo D. A vision-based approach to UAV detection and tracking in cooperative applications[J]. *Sensors*, 2018, 18(10): 3391.
- [4] Shi X, Yang C, Xie W, et al. Anti-drone system with multiple surveillance technologies: Architecture, implementation, and challenges[J]. *IEEE Communications Magazine*, 2018, 56(4): 68-74.
- [5] Sheu B H, Chiu C C, Lu W T, et al. Dual-axis rotary platform with UAV image recognition and tracking[J]. *Microelectronics Reliability*, 2019, 95: 8-17.
- [6] Wang C, Wang T, Wang E, et al. Flying small target detection for anti-UAV based on a Gaussian mixture model in a compressive sensing domain[J]. *Sensors*, 2019, 19(9): 2168.
- [7] Gong M, Guo R, He S, et al. IR radiation characteristics and operating range research for a quad-rotor unmanned aircraft vehicle[J]. *Applied Optics*, 2016, 55(31): 8757-8762.
- [8] Ren S, He K, Girshick R, et al. Faster r-cnn: Towards real-time object detection with region proposal networks[J]. *Advances in neural information processing systems*, 2015, 28.
- [9] Liu W, Anguelov D, Erhan D, et al. Ssd: Single shot multibox detector[C]//*Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I* 14. Springer International Publishing, 2016: 21-37.
- [10] Redmon J, Divvala S, Girshick R, et al. You only look once: Unified, real-time object detection[C]//*Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016: 779-788.
- [11] He K, Gkioxari G, Dollár P, et al. Mask r-cnn[C]//*Proceedings of the IEEE international conference on computer vision*. 2017: 2961-2969.
- [12] Aziz L, Fe M S B H S, Ayub S. Multi-level refinement enriched feature pyramid network for object detection[J]. *Image and Vision Computing*, 2021, 115: 104287.
- [13] Weiguang L I U, Dong L I U, Lu W. Survey of Deformable Convolutional Networks[J]. *Journal of Frontiers of Computer Science & Technology*, 2023, 17(7): 1549.
- [14] Jang B, Kim M, Harerimana G, et al. Bi-LSTM model to increase accuracy in text classification: Combining Word2vec CNN and attention mechanism[J]. *Applied Sciences*, 2020, 10(17): 5841.
- [15] Liu Y, Liu J, Ke Y. A detection and recognition system of pointer meters in substations based on computer vision[J]. *Measurement*, 2020, 152: 107333.
- [16] Everingham M, Eslami S M A, Van Gool L, et al. The pascal visual object classes challenge: A retrospective[J]. *International journal of computer vision*, 2015, 111: 98-136.
- [17] Wang Y, Chen Y, Choi J, et al. Towards visible and thermal drone monitoring with convolutional neural networks[J]. *APSIPA Transactions on Signal and Information Processing*, 2019, 8: e5.